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Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Task Description:	Fluorescence resonance energy transfer (FRET), introduced into biology by Stryer and Hougland, has been extensively used in molecular biological studies. This technique was extended to single molecules by S. Weiss, D. Chemla and colleagues, where FRET changes were seen as freely suspended molecules drifted into a confocal observation. We departed from this pioneering work by studying the structural changes on individual molecules immobilized on a microscope cover slip. In our previous work, we have demonstrated in a wide range of systems that molecules immobilized on properly prepared surfaces can be used to study the enzymology, kinetics, and structural changes of biological molecules. We propose to continue to develop single molecule methods for studying biological processes that will also have wide applicability in single molecule work. The main drive in developing these methods is to answer specific biological questions we have posed. Since we freely share our knowledge with many other groups around the world, the techniques developed has consequences far beyond the work in our lab.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A) Technical advancements
Nanometer colocalization of multi-fluorophores. We have continued to push the limits in the technology we have developed to detect in real-time and space the assembly processes of large multi-component complexes during transcription initiation of RNA Polymerase II. In our scheme a promoter containing DNA is manipulated with an optical trap while fluorescent-labeled protein factors that bind to the DNA are imaged and localized with nanometer precision. Different transcription factors labeled with different color fluorophores are imaged on separate regions of the CCD and thus their relative distances can be resolved beyond the Rayleigh limit. We have substantially improved our technical capabilities in the last 12 months by:

1) Increasing the stability of the microscope mechanical stage to 3Angstrom rms in closed loop mode and reducing the mechanical drift in open loop mode to <1nm/min by stabilizing the temperature of a home-built microscope to <50mK peak-to-peak over >1day.

2) Increasing the registration accuracy of separate color images to 4-5Angstrom rms and inventing a method of absolute calibration of a position measuring instrument to <1A Angstrom rms by combining phase-sensitive detection with feedback stabilization.

3) Optically manipulating a microsphere to clamp its position to 1Angstrom rms using ~1pN forces. On separate experiments we measured the rupture forces for TFIID-DNA complexes to be >10pN. This suggests that optical manipulation could successfully reduce the Brownian motion of these complexes without disrupting their structure.

Optimizing labeling and biochemical assays of the ribosome for single-molecule spectroscopy. For tracking of dynamics in single molecule assays, we dye-labeled tRNAs which proved to work through the initial selection and proofreading scheme. Further to study the dynamics of ribosome itself, we developed a novel method to label ribosomal RNA incorporating mutated hairpins at less conserved helices of the ribosomal RNAs. Combined with another strategy in which we can label ribosomal proteins, we are provided with a wide choice of labeling sites with which we can track the internal dynamics of ribosomes through single molecule FRET measurements. Translation steps like initiation or translocation have been proposed to require the movements of ribosomal units and, thus, with our new labeling techniques, much about the less-known dynamics of ribosome itself would be revealed. Combined with labeled elongation factors, which is in progress, those techniques will also reveal the interaction of elongation factors with ribosome to further elucidate the initial selection and proofreading steps.

Design of microfluidic devices for compartmentalized neuronal culture. We have designed and fabricated microfluidic devices that could culture compartmentalized primary neurons. The fundamental features of the microfluidic device are two chambers inter-connected by micron-size channels. Compartmented culture in microfluidic chambers are maintained under the same condition as in mass cultures. We have successfully cultured compartmented day 0 mouse DRG neurons and hippocampal neurons in the microfluidic device. The viability of DRG neurons inside the microfluidic device is comparable to the mass cultured neurons.

B) Scientific achievements

Ribosome dynamics using single molecule FRET measurements. We used an optical tweezer assay to measure the rupture force between a single ribosome complex and mRNA. It was compared between ribosome complexes assembled on an mRNA with and without a strong Shine–Dalgarno (SD) sequence. Removal of the SD sequence significantly reduced the rupture force, indicating that the SD interaction contributes significantly to the stability of the ribosomal complex on the mRNA before peptide bond formation. By contrast, the presence of a peptidyl tRNA analogue, N-acetyl-Phe-tRNAPhe, in the A site, which mimicked the post-peptidyl transfer state, weakened the rupture force as compared to the complex with Phe-tRNAPhe, and the resultant force was the same for both the SD-containing and SD-deficient mRNAs. These results suggest that formation of the first peptide bond destabilizes the SD interaction, resulting in the weakening of the force with which the ribosome grips an mRNA. This might be an important requirement to facilitate movement of the ribosome along mRNA during the first translocation step.

Single molecule imaging of NGF transport in axons. We used quantum dot-labeled NGF (QD-NGF) to track the movement of NGF in real time to gain insight into the molecular mechanisms of NGF transport. Retrograde axonal transport of Nerve Growth Factor (NGF) signals is critical for the survival, differentiation, and maintenance of peripheral sympathetic and sensory neurons and basal forebrain cholinergic neurons. However, the mechanisms by which the NGF signal is propagated from the axon terminal to the cell body are yet to be fully elucidated. Compartmentalized culture of rat dorsal root ganglion (DRG) neurons was used. Our studies showed that active transport of NGF within the axons was characterized by rapid, unidirectional movements interrupted by frequent pauses. Almost all movements were retrograde, but short-distance anterograde movements were occasionally observed. Surprisingly, quantitative analysis at the single molecule level demonstrated that the majority of NGF-containing endosomes contained only a single NGF dimer. Electron microscopic analysis of axonal vesicles carrying QD-NGF confirmed this finding. The majority of QD-NGF was found to localize in vesicles 50-150 nm in diameter with a single lumen and no visible intralumenal membranous components. Our findings point to the possibility that a single NGF dimer is sufficient to sustain signaling during retrograde axonal transport to the cell body.

Single-molecule studies of the Eykaryotic transcriptional machinery. We have produced various fluorescent versions of recombinant general transcription factors TFIIB-TFIIA-TFIIF and have site-specific labeled with high efficiency (~70%) native TFIID using Quantum dots and a biotinylated monoclonal antibody against the TFIID subunit TAF250. This is from the first examples of successful labeling of a native ~1MDa multi-subunit complex for single molecule studies. We have shown that the Qdot labeled TFIID binds to surface immobilized promoter DNA in a DNA sequence specific manner and causes a conformational change on the DNA consistent with wrapping of an extended DNA region around the protein complex. Furthermore we observe in preliminary single molecule experiments that TFIIA increases the stability of the TFIID-DNA complex whereas TFIIB has no effect on TFIID-promoter recognition and enters the Pre-initiation complex assembly at a later stage. Finally, in single molecule experiments the purified TFIID complex is efficiently recruited to surface tethered DNA by the synthetic activator Gal4-VP16 and the measured DNA end-to-end distance is consistent with a DNA loop forming between the upstream bound activator and the promoter bound TFIID

	complex. Combined single molecule force and fluorescence microscope to control and monitor macromolecular function. Using the single molecule force and fluorescence imaging techniques developed in the lab, we measured the pairwise interactions between cadherin monomers and cis dimers that were formed by fusing the protein with the dimerized Fc region of human IgG. Using single molecule AFM force spectroscopy we determined that lateral dimers are not required for trans adhesion. In fact adhesive complexes formed by the binding of cadherin monomers display a remarkable range of rupture forces which are titrated by the formation of cis dimers. This modulation of adhesion suggests multiple mechanical functions for cadherin, including transient recognition as well as stable interactions in tissue formation. To identify cadherin orientation within dimers, we introduced cysteine residues in different regions of the protein, specifically labeled them with fluorescent dyes and crosslinked the labeled protein under different calcium conditions to capture the low affinity cis dimers. We determined the protein orientation within the dimer molecules using single molecule FRET, single-molecule high-resolution imaging with photobleaching (SHRImP) and single-molecule high-resolution colocalization (SHREC).
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